

Research paper

Preventing depression in Chinese children and adolescents: A pilot study of a brief school-based cognitive behavioral group program

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ABSTRACT

Purpose: Depressive symptoms are common among children and adolescents in China, yet few rigorously evaluated, evidence-based prevention programs have been developed. This pilot study evaluated the feasibility, acceptability, and preliminary efficacy of Power Up: Cognitive Behavioral Therapy for Depressive Symptoms (Power Up-CBT), a brief, manualized group intervention tailored for Chinese school-aged youth with depressive symptoms.

Methods: Eighty-seven students from grades 5, 6, and 9 who screened positive for depressive symptoms were randomly allocated to Power Up-CBT or treatment-as-usual (TAU). Feasibility was assessed via attrition, attendance, satisfaction, and qualitative feedback. Symptoms of depression and anxiety were evaluated at baseline and post-intervention using the Center for Epidemiologic Studies Depression Scale-Revised (CESD-R), the Children's Depression Inventory-Short Form (CDI-S), and the Screen for Child Anxiety Related Emotional Disorders (SCARED).

Results: Among forty-three participants in the Power Up-CBT group, one withdrew (attrition rate = 2.3 %), and mean attendance was 87.7 %. Satisfaction was reasonable ($M = 28.4/32$, $SD = 4.06$). Qualitative feedback reflected improved coping, positive group experiences, and interest in additional sessions.

At post-intervention, depressive symptoms decreased significantly more in the Power Up-CBT group than in TAU (CESD-R adjusted mean difference [AMD] = -10.89 , $p < .001$; CDI-S AMD = -3.23 , $p < .001$). While anxiety symptoms also declined, the between-group difference did not reach statistical significance (AMD = -6.68 , $p = .07$).

Conclusion: Power Up-CBT demonstrated reasonable feasibility and acceptability, showing promising preliminary effects in reducing depressive symptoms among school-aged youth in mainland China. Larger randomized trials with extended follow-up are warranted.

1. Introduction

Depression ranks among the most prevalent mental health challenges facing young populations (Shorey et al., 2022) and is anticipated to be a major contributor to global disease burden by 2030 (Malhi and Mann, 2018). In China, approximately 1.3 % of children and adolescents meet criteria for Major Depressive Disorder (Xu et al., 2018), while sub-threshold depressive symptoms are even more widespread, affecting approximately 19.85 % (Rao et al., 2019). Yu and Huang (2023) further reported prevalence rates of 13.5 % in primary school students, 23.9 % in junior middle school, and 28.0 % in senior high school. Early-onset depression is associated with cognitive impairments, emotional

instability, health issues, and social withdrawal (Bang et al., 2012; Gleason and Thompson, 2022; Jiang et al., 2021), which may lead to impaired neurodevelopment, low self-esteem, academic difficulties, suicidal ideation, and heightened risk for future psychiatric conditions (Eyre and Thapar, 2014; Gray et al., 2024; Izaki, 2021; Moukaddam et al., 2019).

Schools play a vital role in the early identification and prevention of depression among children and adolescents (Lai et al., 2022; Werner-Seidler et al., 2021). Nonetheless, the ongoing lack of mental health personnel in China limits the scalability of traditional, clinician-led approaches, which are often overly complex and difficult to implement across school systems (Yang et al., 2023). Although national

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strategies like the Healthy China 2030 Plan, the Children and Adolescents Mental Health Action Plan (2019–2022), and the Special Action Plan for Student Mental Health (2023–2025) have been introduced, access to mental health services within schools remains inadequate, especially in rural areas with insufficient infrastructure and trained personnel (Xiang et al., 2018; Yuan et al., 2024). A major unmet need lies in the absence of brief, empirically supported programs that can be delivered by professionals and general school staff alike. There is an urgent call for scalable and contextually adaptable interventions to alleviate depressive symptoms among young people in mainland China.

Cognitive Behavioral Therapy (CBT) has become a foundational approach in efforts to prevent depression within school contexts. Substantial empirical evidence supports its efficacy as one of the most robust therapeutic strategies for addressing depressive symptoms in both children and adolescents (Clarke et al., 2001; Kambara and Kira, 2021; Keles and Idsoe, 2018; Rosselló et al., 2008; Spirito et al., 2011). It operates on the principle that modifying dysfunctional thoughts can improve maladaptive emotions and behaviors (Crowe and McKay, 2017), and typically includes psychoeducational guidance, behavioral activation, cognitive restructuring, structured problem solving, and auxiliary techniques such as relaxation practices or interpersonal skill-building (American Psychological Association, 2019; Garber et al., 2016b; McCarty and Weisz, 2007; van den Heuvel et al., 2019; Weersing et al., 2009). A meta-regression showed that indicated CBT prevention programs can reduce later depression risk by 63 %, especially when combining behavioral activation and cognitive restructuring (Oud et al., 2019).

Among delivery formats, Cognitive Behavioral Group Therapy (CBGT) is well-suited for school settings (Feiss et al., 2019). It is more time- and cost-efficient than individual therapy (Lewinsohn and Clarke, 1999; Vos et al., 2005), and its group format enhances peer support and engagement (Tian et al., 2013). By reaching multiple students at once, CBGT also mitigates the shortage of school-based mental health staff, making it a scalable and feasible intervention model (Christner et al., 2007; Fu et al., 2020). Beyond delivery format, dosage is critical. Youth programs typically include 8–16 sessions: for example, the Adolescent Coping with Depression Course (ACDC) comprises ten weekly 120-min sessions plus two follow-ups; the Penn Resiliency Program (PRP) offers 10–12 sessions with six booster sessions; Interpersonal Psychotherapy for Depressed Adolescents (IPT-A) spans 12–20 sessions; Op Volle Kracht 2.0 (OVK 2.0) includes eight sessions; Behavioral Activation (BA) often involves 16–20 sessions, whereas brief BA uses eight; and MICHI, a brief outpatient program, consists of six sessions of 75–90 min each (Bernaras et al., 2019; de Jonge-Heesen et al., 2020; Gillham et al., 2012; Gillham et al., 2007; Gunlicks-Stoessel et al., 2010; Idsoe and Keles, 2016; Idsoe et al., 2019; Mufson and Sills, 2006; Pass et al., 2018). Shorter programs are generally associated with better engagement and adherence, whereas longer interventions tend to have higher dropout rates (Dunn et al., 2012; Stice et al., 2009). “Short,” however, should denote a minimal multi-session dose; single-session interventions (SSIs) are intentionally ultra-brief and low-intensity, useful for reach and completion yet typically yielding smaller and less durable effects, and are best positioned as adjuncts rather than substitutes for multi-session CBT (Cohen et al., 2024; Schleider et al., 2020; Schleider et al., 2022; Schleider and Weisz, 2017; Schleider et al., 2025).

Despite the recognized effectiveness of CBT-based approaches for preventing depression in children and adolescents, most studies have been conducted in Western settings, with limited empirical evidence available from mainland China (Duagi et al., 2024; Feiss et al., 2019; Kambara and Kira, 2021; Nardi et al., 2017; Oud et al., 2019). X. Tang et al. (2024a), for instance, implemented a universal school-based CBT prevention program and reported favorable outcomes. However, existing evidence indicates that targeted programs for risk individuals tend to yield stronger effects than those aimed at the universal population (Kambara and Kira, 2021; Rasing et al., 2017; Werner-Seidler et al., 2021). While few culturally adapted CBT programs have shown

promise—such as Ip et al.'s (2016) internet-based intervention for adolescents in Hong Kong—manualized, school-based group CBT interventions specifically designed for children and adolescents with depressive symptoms remain absent in mainland China. Although internet-based CBT (iCBT) offers flexibility and convenience, a school-based, face-to-face format is more feasible and contextually appropriate in mainland China. In-person delivery reduces reliance on home/school devices and internet access, addressing access inequities, particularly in rural areas with limited digital resources (Qin et al., 2024; L. Tang et al., 2024b; Zhang et al., 2025). Additionally, a low-data, in-person approach alleviates concerns about online privacy and avoids conflicts with school policies that restrict smartphone use (General Office of the Ministry of Education of the People's Republic of China, 2021; Zhang et al., 2025).

International experience indicates that school-based CBT is feasible and acceptable in educational settings (Arora et al., 2019; Feiss et al., 2019; Kambara and Kira, 2021; Michael et al., 2016). For example, a school-based cognitive-behavioral skills-building group program has shown good feasibility and positive participant feedback (Mazurek Melnyk et al., 2014). Emerging evidence in mainland China supports the feasibility of school-based CBT, including universal and trauma-focused CBT programs, suggesting that structured psychological interventions can be implemented within Chinese school systems (Li et al., 2023; X. Tang et al., 2024a).

To fill the gap in existing programs and ensure both contextual relevance and practical feasibility in under-resourced school settings, we developed Power Up: Cognitive Behavioral Therapy for Depressive Symptoms (Power Up-CBT), a brief, manualized group intervention grounded in evidence-based CBT prevention models and implementation research principles. The Consolidated Framework for Implementation Research (CFIR; Damschroder et al., 2009; Damschroder et al., 2022) was used to guide early-stage design considerations, including intervention design, cultural adaptability, delivery mechanisms, and contextual fit within school settings. This pilot randomized controlled trial (RCT) aimed to evaluate the feasibility, acceptability, and preliminary effectiveness of Power Up-CBT. It was hypothesized that Power Up-CBT would demonstrate good feasibility and acceptability and yield greater reductions in depressive symptoms than treatment-as-usual (TAU). Given the frequent co-occurrence of depression and anxiety in youth and CBT used in Power Up-CBT is evidence-based for preventing both conditions, we pre-specified anxiety as a secondary outcome to evaluate potential spillover effects of a disorder-specific program, and hypothesized greater reductions in anxiety symptoms among Power Up-CBT participants versus TAU (Cummings et al., 2014; Garber et al., 2016a).

2. Methods

2.1. Trial design

A randomized controlled trial was implemented. To mitigate assessment-related bias, data collectors were blinded to the participants' allocation. Ethical clearance was granted by the Ethics Committee of the School of Social Development and Public Policy at Beijing Normal University (Approval No. SSDPP-HSC-20230015), and the trial was formally recorded in the Chinese Clinical Trial Registry (ChiCTR2300076462). Informed consent was obtained from all participating students and their legal guardians.

2.2. Participants

The study was conducted in two elementary schools and one junior middle school in Pingdingshan, Henan Province, China. Following consultation with the schools, Grades 5, 6, and 9 were selected on scheduling grounds (e.g., availability of after-class sessions). With support from school staff, students in these grades, together with their

guardians, were briefed on the purpose of the study and invited to join on a voluntary basis. Class-based screening procedures were implemented, and students who met preliminary criteria received additional information about the intervention and study protocol. Students who expressed interest submitted signed informed consent forms, duly completed by both the student and their legal guardian, within the specified enrollment window.

Eligibility criteria for participation included: (a) being 10–15 years old; (b) scoring 16 or above on the Center for Epidemiologic Studies Depression Scale-Revised (CESD-R); and (c) demonstrating adequate language comprehension and expression (written or spoken). Participants were excluded if caregivers or school staff reported: (a) serious developmental delays or acute psychiatric conditions requiring immediate medical attention; (b) a heightened risk of suicide (e.g., recent suicidal plans, or attempts); or (c) ongoing mental health treatment beyond standard school-based support. These criteria ensured participants' safety and appropriate care, as students at heightened suicide risk require immediate, individualized clinical attention, which is outside the scope of this group-based intervention. Monitoring procedures involved regular communication with school staff or caregivers to identify exclusion criteria. Any such student was not enrolled/withdrawn and received same-day referral to designated school mental health staff or the homeroom teacher for parent/guardian notification, safety planning, and clinical referral as indicated.

Data were collected at two time points: a baseline assessment (T0) and a post-intervention follow-up (T1), which was completed within two weeks of the final group session.

2.3. Sample size

According to a prior meta-analysis, for children and adolescents with (subclinical) depression have demonstrated moderate effectiveness, reporting a standardized mean difference (SMD) of -0.41 (Oud et al., 2019). Based on this estimate, a power analysis using G*Power 3.1 (Faul et al., 2007) indicated that detecting a similar effect at 80 % power and $\alpha = 0.05$ would require approximately 95 participants per group. After accounting for an anticipated 10 % attrition rate, a fully powered randomized controlled trial would require about 212 participants in total. Considering that this pilot study primarily focused on evaluating feasibility and acceptability, we targeted approximately 20 % of that sample size, yielding an intended total sample of 42 participants.

2.4. Randomization and masking

Eligible participants were assigned unique anonymized identification (ID) codes in the order eligibility was confirmed. IDs were stored separately from personally identifiable information to maintain confidentiality. The anonymized ID list was provided to an independent research assistant responsible for randomization.

Given the staggered program initiation across participating schools, randomization was stratified by school. Within each school, participants were randomly allocated approximately 1:1 to either the Power Up-CBT group or the TAU condition based on a computerized randomization sequence.

2.5. Interventions

2.5.1. The Power Up-CBT group

Participants in the Power Up-CBT group received a structured, manualized cognitive behavioral group intervention developed by researchers at Beijing Normal University. Informed by extensive reviews of evidence-based CBT prevention and treatment programs for children and adolescents (e.g., Briere et al., 2019; de Jonge-Heesen et al., 2020; Gillham et al., 2007; Oud et al., 2019; Pössel et al., 2004; Singhal et al., 2018; van den Heuvel et al., 2019; Wijnhoven et al., 2014), the program integrated five core CBT components: (1) psychoeducation, (2)

behavioral activation, (3) problem-solving, (4) cognitive restructuring, and (5) relapse prevention.

In line with CFIR's emphasis on cultural appropriateness and user-centered design, Power Up-CBT incorporated adaptations tailored to the psychological development and sociocultural context of Chinese children and adolescents. In China, academic achievement is closely tied to self-worth, family honor, and social status. Traditional values—such as *wan ban jie xia pin*, *wei you du shu gao* (“all pursuits are inferior to study”) and *wang zi cheng long* (“high parental expectations”)—contribute to an intensely competitive educational environment. These cultural pressures, including academic overload, internalized perfectionism, and performance-based self-esteem, have been linked to elevated depression risk among students (Quach et al., 2015; Zhou et al., 2021; Zhou et al., 2024).

For example, to ensure relevance, the behavioral activation module encouraged students to reflect on their emotional responses to academic setbacks (e.g., poor test performance) and to engage in culturally resonant activities such as the “Egg–Chick–Phoenix” game—a metaphor for transformation and growth through adversity. These experiential exercises aimed to redirect attention from negative self-evaluation to positive emotional experiences. Students also generated individualized “pleasure lists”, which were broken into actionable steps to promote sustained engagement in enjoyable and rewarding activities.

In the cognitive restructuring module, the program introduced visually engaging “Thought Monsters”—metaphorical characters representing common cognitive distortions such as catastrophizing and self-blame (Leahy, 1988). Each monster was illustrated as a cartoon figure with features that reflected its specific distortion, enhancing visual appeal and symbolic clarity. To further enhance relatability and conceptual accessibility, each monster was given unique catchphrases drawn from language that children and adolescents commonly use and understand. These phrases reflected the monster's specific cognitive distortion, making it easier for students to recognize and label these patterns. Through interactive games such as “Beat the Monster,” students practiced identifying and challenging maladaptive thought patterns in a playful and developmentally appropriate format.

The program consisted of seven weekly sessions (50–60 min each), following a standardized structure: (1) homework review, (2) introduction of new content, and (3) assignment of take-home practice. Age-appropriate materials and strategies were employed to sustain engagement (Garber et al., 2016b), including:

Role-playing and storytelling

Group-based experiential games

Visual metaphors and mood check-ins (e.g., 1–10 emotion rating scales)

The content of each session is outlined in Supplementary Table S1.

The intervention was implemented in a group setting by four research-team trained facilitators, including one PhD student with expertise in child and adolescent mental health, alongside three master's-level students in social work. Two served as lead facilitators, while the others supported as co-facilitators. All facilitators underwent a structured three-day training and received weekly supervision from a senior CBT-certified clinician. Sessions were scheduled during non-academic hours in coordination with school staff to minimize disruption to classroom learning. Weekly team debriefings were conducted to monitor implementation fidelity, assess delivery context, and guide responsive adaptations throughout the intervention.

Collectively, these design and delivery components reflect structured implementation planning aligned with CFIR's emphasis on innovation attributes, team formation, facilitator training, and adaptation to local conditions.

2.6. The TAU group

Under the standard educational curriculum, children and adolescents in the TAU group participated in regularly scheduled classes on

ideological and moral education as well as general mental health education. These sessions typically emphasized values-based learning, emotional awareness, and basic coping strategies, and were embedded into the schools' regular academic schedules. In addition to these structured lessons, students in the TAU group also had access to routine psychological services provided by the school, based on individual needs. At the conclusion of the study, students in the TAU group who were identified as still requiring additional support were offered access to the intervention, which was delivered by school teachers trained by the program team.

2.7. Measures

2.7.1. Center for Epidemiologic Studies Depression Scale-Revised (CESD-R)

The CESD-R is a 20-item self-administered scale used to assess the severity of depressive symptoms in accordance with DSM criteria (Eaton et al., 2004; Radloff, 1977). Scores span from 0 to 60, with elevated scores reflecting more intense depressive symptomatology. This tool has demonstrated high internal reliability among Chinese youth populations (Cronbach's $\alpha = 0.82\text{--}0.96$) and is widely recognized as a suitable screening measure for children and adolescents (Ip et al., 2016; Shek et al., 2022). In the current study, the CESD-R served as the primary screening tool for identifying participants with depressive symptoms.

2.7.2. Children's Depression Inventory-Short Form (CDI-S)

The CDI-S is a brief, 10-item questionnaire designed to identify depressive symptoms in youths aged 7–17 years (Carey et al., 1987). Each item presents three response choices reflecting symptom frequency, with response values ranging from 0 to 2. Higher total scores indicate more pronounced depressive symptomatology. This tool has been widely employed in Chinese samples and has demonstrated solid psychometric properties, with a reported Cronbach's α coefficient of 0.84 (de la Vega et al., 2016).

2.7.3. Screen for Child Anxiety Related Emotional Disorders (SCARED)

The SCARED consists of 41 items and is a self-administered questionnaire designed to evaluate anxiety-related symptoms in youth populations (Birmaher et al., 1997). Respondents rate each item on a 3-point Likert scale ranging from 0 to 2, with scores summed to yield a maximum of 82 points. The Chinese adaptation has demonstrated satisfactory psychometric properties, including acceptable reliability indices in school-aged samples (Wang et al., 2002).

2.7.4. Participant adherence

Participant adherence was assessed based on session attendance and post-intervention evaluations. Attendance was recorded by a trained research assistant using a standardized coding system.

2.7.5. Participant satisfaction

To evaluate participants' satisfaction with the intervention, the Client Satisfaction Questionnaire-8 (Kelly et al., 2018) was administered. This measure includes 8 questions, each answered on a 4-point Likert scale ranging from 1 (extremely dissatisfied) to 4 (extremely satisfied), producing total scores from 8 to 32. Based on total scores, satisfaction was categorized as low (8–20), moderate (21–26), or high (27–32) (Attkisson and Zwick, 1982).

2.7.6. Participant acceptability

Participants' perceptions of acceptability were assessed through qualitative feedback via open-ended questions administered by a research assistant after the final session. Interviews were conducted separately within each intervention group in a brief semi-structured format, using neutral probes as needed. To minimize bias, facilitators were not present during assessments. Participants were asked to share feedback on various aspects of the program, such as their experiences with therapy sessions, homework tasks, and whether the intervention

addressed their concerns (e.g., “What are your overall impressions?”, “What changes have you noticed?”, “Have you learned new ways to manage your feelings?”, “What would you improve?”, “Is there anything else you'd like to share?”).

2.8. Data analysis

All analyses followed an intent-to-treat (ITT) approach. Missing data were addressed using single imputation with a non-parametric random forest algorithm (missForest), except for satisfaction data, which were examined solely among those who finished the post-intervention survey.

Baseline demographic and clinical characteristics of participants from the Power Up-CBTD and TAU groups were described using summary statistics. Group comparisons were conducted via chi-square tests for categorical data and independent *t*-tests for continuous measures.

To assess intervention effects on depressive and anxiety symptoms, analysis of covariance (ANCOVA) was employed using ordinary least squares regression. CESD-R, CDI-S, and SCARED scores at posttest served as outcome variables, with group allocation as the fixed factor and baseline scores as covariates.

Additional sensitivity analyses were conducted by including gender as a covariate and using complete-case data (participants with no missing outcomes). All statistical procedures were carried out in R (version 4.4.2) and Python (version 3.13).

3. Results

3.1. Participants' characteristics

Fig. 1 presents the CONSORT flowchart illustrating the progression of participants throughout the study. A total of 1960 students were screened, and 87 who met the inclusion criteria and provided signed consent were enrolled in the study. Informed consent was obtained within a limited enrollment window, constrained by factors such as fixed school schedules and a predefined study timeline. One participant in the Power Up-CBTD group withdrew after the first session. In total, 84 participants (96.6 %) completed the post-intervention assessment.

According to Table 1, a total of 87 children and adolescents participated ($M = 12.47$ years, $SD = 1.70$), comprising 35 boys (40.2 %) and 52 girls (59.8 %). Participants were mainly from grades 5, 6, and 9, and the majority lived in urban areas and came from married families. Most had siblings and were non-boarders. The Power Up-CBTD ($n = 43$) and TAU ($n = 44$) groups were comparable in most demographic characteristics (e.g., age, grade, region, family structure; $ps > 0.05$), except for gender ($p < .05$).

3.2. Feasibility implementation outcomes

3.2.1. Participant adherence

Among the 43 participants assigned to the Power Up-CBTD group, one participant discontinued after the initial session (attrition rate = 2.3 %), and two missed the post-intervention assessment due to illness. Consequently, adherence analysis was conducted on 42 participants. The intervention was initially designed as a seven-session group-based program. However, due to scheduling constraints on the day of the sixth session and the density of its content, three groups extended the intervention to eight sessions, while one group followed the original seven-session schedule. The overall attendance rate was 87.7 % (285 out of 325 sessions attended). At the individual level, 54.8 % of participants attended all sessions, 21.4 % missed one.

3.2.2. Participant satisfaction

Of the 43 children and adolescents who participated in the Power Up-CBTD group, 40 (93.0 %) completed the post-intervention satisfaction survey. Participants indicated strong satisfaction with the Power Up-CBTD intervention, reflected in an average rating of 28.4 ($SD = 4.06$)

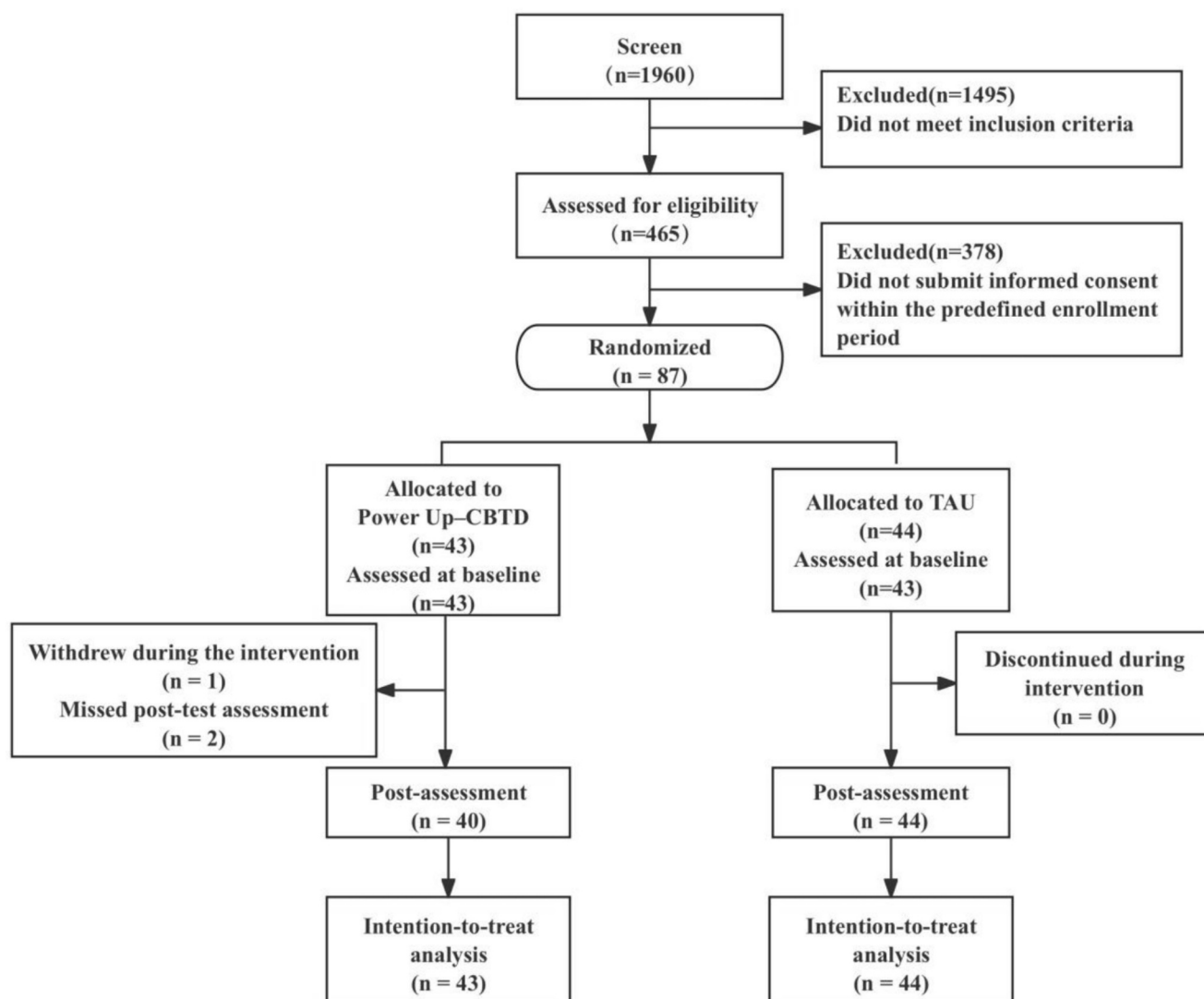


Fig. 1. CONSORT flowchart.

Notes: Power Up-CBT = power up: cognitive behavioral therapy for depressive symptoms; TAU = treatment as usual.

on the satisfaction scale. Notably, 67.5 % ($n = 28$) indicated high satisfaction, and an additional 22.0 % ($n = 9$) reported moderate satisfaction.

Regarding specific evaluation items, 97.5 % of participants expressed satisfaction with the overall quality of the group services, and 97.5 % agreed that the content provided met their individual needs. Additionally, 95.0 % reported that the intervention helped alleviate their current difficulties. Furthermore, 97.5 % expressed overall satisfaction with the group experience, and 92.5 % indicated a willingness to participate in similar group services in the future. In response to the supplementary question, “If you had known about the group service earlier, would you have joined sooner?”, 90.0 % responded affirmatively.

3.2.3. Participant acceptability

Children and adolescents who participated in the Power Up-CBT intervention generally reported substantial gains, particularly in coping skills and emotional awareness. For example, some shared that visualizing negative thoughts as “little monsters” made it easier for them to recognize and challenge unhelpful thinking patterns. Others said that breaking down complex tasks into smaller, manageable steps helped them overcome feelings of helplessness and enabled them to complete tasks they had previously avoided. One participant reflected, “I used to feel that many things were just too hard, even to start. But now, by

breaking things down, I’ve realized I can gradually work through them.”

Participants also gave highly positive evaluations of both the group atmosphere and session format. They described the environment as caring, equal, and free from academic pressure, which made it easier to open up and express themselves. Many reported feelings supported by both peers and facilitators, contributing to a strong sense of emotional safety. One participant shared, “When I talked about things from the past in the group, I received a lot of care, which made me feel warm.”

The sessions were frequently described as engaging and enjoyable. Participants appreciated the interactive format and the absence of exams or written homework. Instead, learning was embedded in life-based exploration and practical tasks. As one participant noted, “There were no tests or paper homework in the group. It was all interesting life exploration and practical tasks. I really liked this format.” Several participants also expressed a wish for a longer program, saying they found the weekly sessions too short.

In terms of perceived impact, many participants reported applying newly learned skills in daily life. Some described increased social engagement and adopting a more optimistic mindset when encountering difficulties. Others reported better emotional regulation, such as taking breaks and seeking support instead of becoming overwhelmed. Several also shared cognitive shifts in balancing academic responsibilities with personal well-being. As one participant said, “I’m about to take the high

Table 1

Demographic characteristics of participants in the Power Up-CBT and TAU groups.

Characteristic		Total (N = 87)	Power Up-CBT (N = 43)	TAU (N = 44)
Age, mean (SD)		12.47 (1.7)	12.56(1.76)	12.38 (1.66)
Gender	Boy	35(40.2 %)	24(55.8 %)	11(25.0 %)
	Girl	52(59.8 %)	19(44.2 %)	33(75.0 %)
School grade	5	18(20.6 %)	10(23.3 %)	8(18.1 %)
	6	27(31.0 %)	12(27.9 %)	15(34.0 %)
	9	42(48.4 %)	21(48.8 %)	21(47.9 %)
Region	Urban	77(88.5 %)	38(88.3 %)	39(88.6 %)
	Rural	10(11.5 %)	5(11.7 %)	5(11.4 %)
Parent's marital status	Married	66(75.9 %)	31(72.0 %)	35(79.5 %)
	Remarried	15(17.2 %)	9(20.9 %)	6(13.6 %)
Child status	Single	6(6.9 %)	3(7.1 %)	3(6.9 %)
	Has siblings	80(92.0 %)	40(93.0 %)	40(90.9 %)
	Only child	7(8.0 %)	3(7.0 %)	4(9.1 %)
Boarders (living in school)	Yes	1(1.1 %)	0	1 (2.2 %)
	No	86(98.9 %)	43(100 %)	43(97.8 %)

Notes: POWER UP-CBT = power up: cognitive behavioral therapy for depressive symptoms; TAU = treatment as usual. Data are shown as n (%) unless stated otherwise.

school entrance exam, so why should I keep suppressing myself? As long as I maintain my grades and have fun, it's fine."

Finally, participants reported improvements in social functioning and personality traits. Many said they became more comfortable initiating conversations and described themselves as more outgoing. "Recently, I've been doing much better. I can talk to my classmates naturally now, and I've become more extroverted," one participant shared. Others noted that they began incorporating more enjoyable activities into their routines and reduced the pressure they placed on themselves. One summarized, "Now I do things that make me happy during breaks from studying. I've learned to ease the pressure I place on myself."

3.3. Treatment outcomes

3.3.1. Depression severity

From Table 2 and Fig. 2, the Power Up-CBT group exhibited significantly lower CESD-R scores at posttreatment compared to the TAU

Table 2

Outcomes at baseline and posttreatment.

	Power Up-CBT			TAU			OLS regression analysis	p-Value	Effect size (d)
	Mean	SD	95 % CI	Mean	SD	95 % CI	Adjusted mean difference (95 % CI)		
CESD-R depression									
Baseline	30.60	10.19	27.47–33.74	31.02	9.42	28.16–33.89			
Posttreatment	20.14	15.26	15.44–24.84	31.25	13.10	27.27–35.23	–10.89 (–16.56 to –5.22)	<0.001	–0.78
CDI-S depression									
Baseline	9.56	3.91	8.36–10.76	9.64	3.42	8.60–10.68			
Posttreatment	6.26	5.09	4.69–7.82	9.55	4.16	8.28–10.81	–3.23 (–4.85 to –1.61)	<0.001	–0.71
SCARED anxiety									
Baseline	37.72	15.06	33.09–42.36	38.23	14.01	33.97–42.49			
Posttreatment	31.56	21.89	24.82–38.29	38.52	15.24	33.89–43.16	–6.68 (–13.96–0.60)	0.07	–0.37

Notes: CESD-R = Center for Epidemiologic Studies Depression Scale-Revised; CDI-S = Children's Depression Inventory-Short Form; SCARED = Screen for Child Anxiety Related Emotional Disorders; POWER UP-CBT = power up: cognitive behavioral therapy for depressive symptoms; TAU = treatment as usual.

group (mean [SD], 20.14 [15.26] vs. 31.25 [13.10]; adjusted mean difference [AMD], –10.89; 95 % CI [–16.56, –5.22]; $p < .001$). This represents a 34.2 % decrease in depressive symptoms in the Power Up-CBT group (from 30.60 to 20.14), whereas the TAU group showed a negligible change of 0.61 % (from 31.06 to 31.25).

CDI-S scores followed a similar pattern, as illustrated in Table 2 and Fig. 3. The posttreatment mean score in the Power Up-CBT group was significantly lower than that in the TAU group (mean [SD], 6.26 [5.09] vs. 9.55 [4.16]; AMD, –3.23; 95 % CI [–4.85, –1.61]; $p < .001$). This reflects a 34.51 % reduction in depressive symptoms in the Power Up-CBT group (with scores decreasing from 9.56 to 6.26), while the TAU group showed virtually no change (decreasing slightly from 9.64 to 9.55).

3.3.2. Anxiety severity

Regarding anxiety symptoms, Table 2 and Fig. 4 indicate that the Power Up-CBT group exhibited lower posttreatment SCARED scores compared to the TAU group (mean [SD], 31.56 [21.89] vs. 38.52 [15.24]; AMD, –6.68; 95 % CI [–13.96, 0.60]; $p = .07$). Although the between-group difference was not statistically significant, the intervention group exhibited a 16.32 % reduction in anxiety symptoms (with scores decreasing from 37.72 to 31.56), whereas the TAU group showed a slight increase (from 38.23 to 38.52).

3.3.3. Sensitivity analyses

To account for baseline gender imbalance, we conducted a sensitivity analysis including group $\times$ gender interaction terms in the OLS models. The interaction effects were nonsignificant across all outcomes—CESD-R Depression ($p = .806$), CDI-S Depression ($p = .986$), and SCARED Anxiety ($p = .568$)—indicating that the intervention effects did not differ between boys and girls.

In addition, complete-case analyses were conducted using only participants with no missing data on each respective scale. Results were consistent with those of the main analyses. For CESD-R ($n = 72$), the AMD was –11.29 (95 % CI [–17.77, –4.87], $p = .001$); for CDI-S ($n = 71$), the AMD was –2.99 (95 % CI [–4.77, –1.22], $p = .001$); and for SCARED ($n = 59$), the AMD was –6.68 ($p = .11$).

4. Discussion

This pilot study presents the first empirical evaluation in mainland China of a brief, manualized, cognitive behavioral group program delivered in school settings, tailored for children and adolescents experiencing depressive symptoms. Power Up-CBT was developed in response to the growing mental health needs among Chinese children and adolescents and the lack of culturally relevant, evidence-based prevention models in the region.

The intervention showed promising feasibility and acceptability. Attrition was low, with only one participant withdrawing after the first session (2.3 %), and the average attendance rate was 87.7 %, indicating

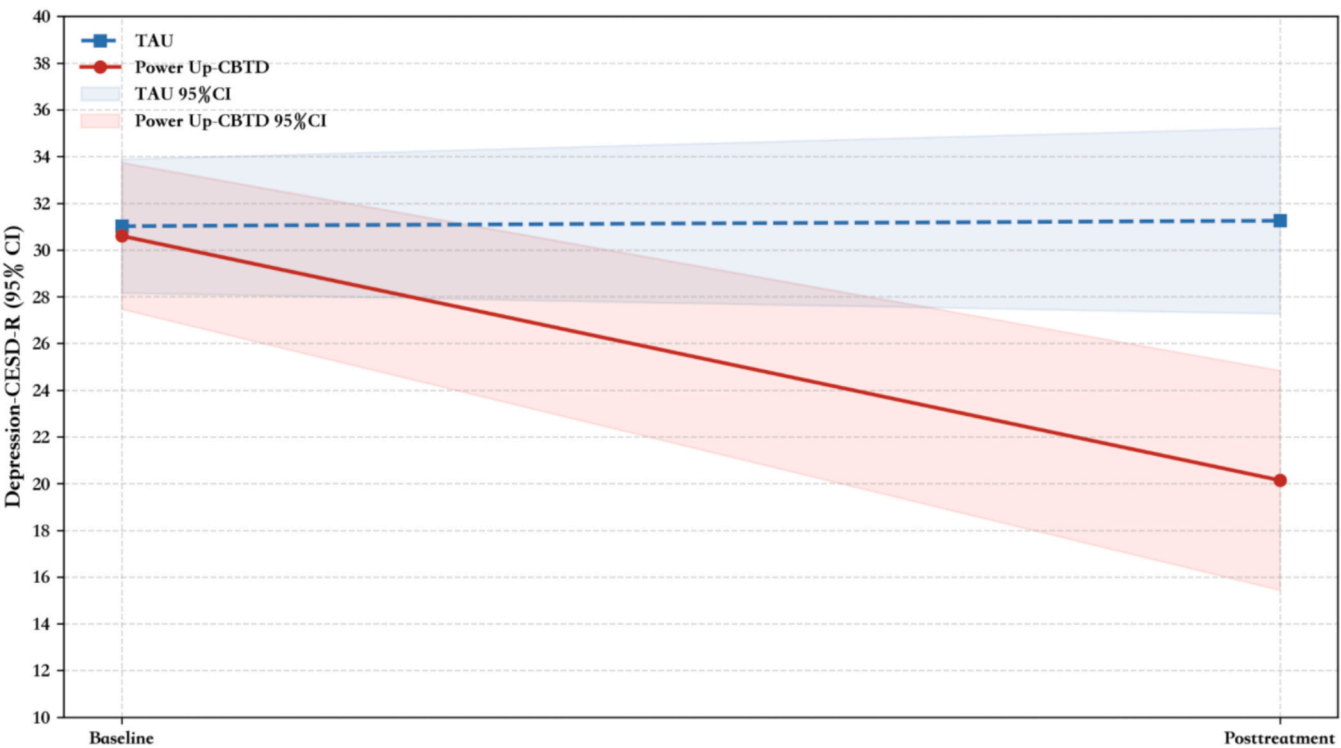


Fig. 2. Effects of the power up-CBTD versus TAU on CESD-R scores.
Notes: CESD-R = Center for Epidemiologic Studies Depression Scale-Revised; Power Up-CBTD = power up: cognitive behavioral therapy for depressive symptoms; TAU = treatment as usual.

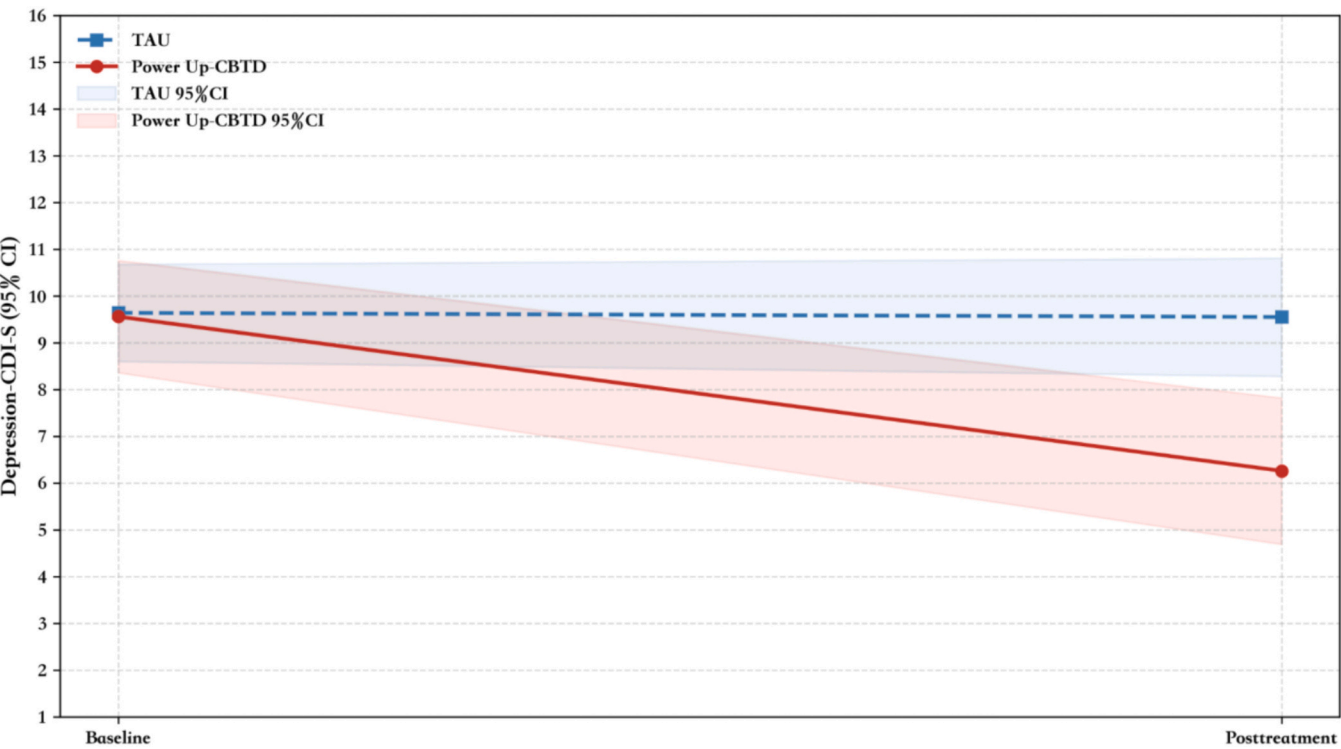


Fig. 3. Effects of the Power Up-CBTD versus TAU on CDI-S scores.
Notes: CDI-S = Children's Depression Inventory-Short Form; Power Up-CBTD = power up: cognitive behavioral therapy for depressive symptoms; TAU = treatment as usual.

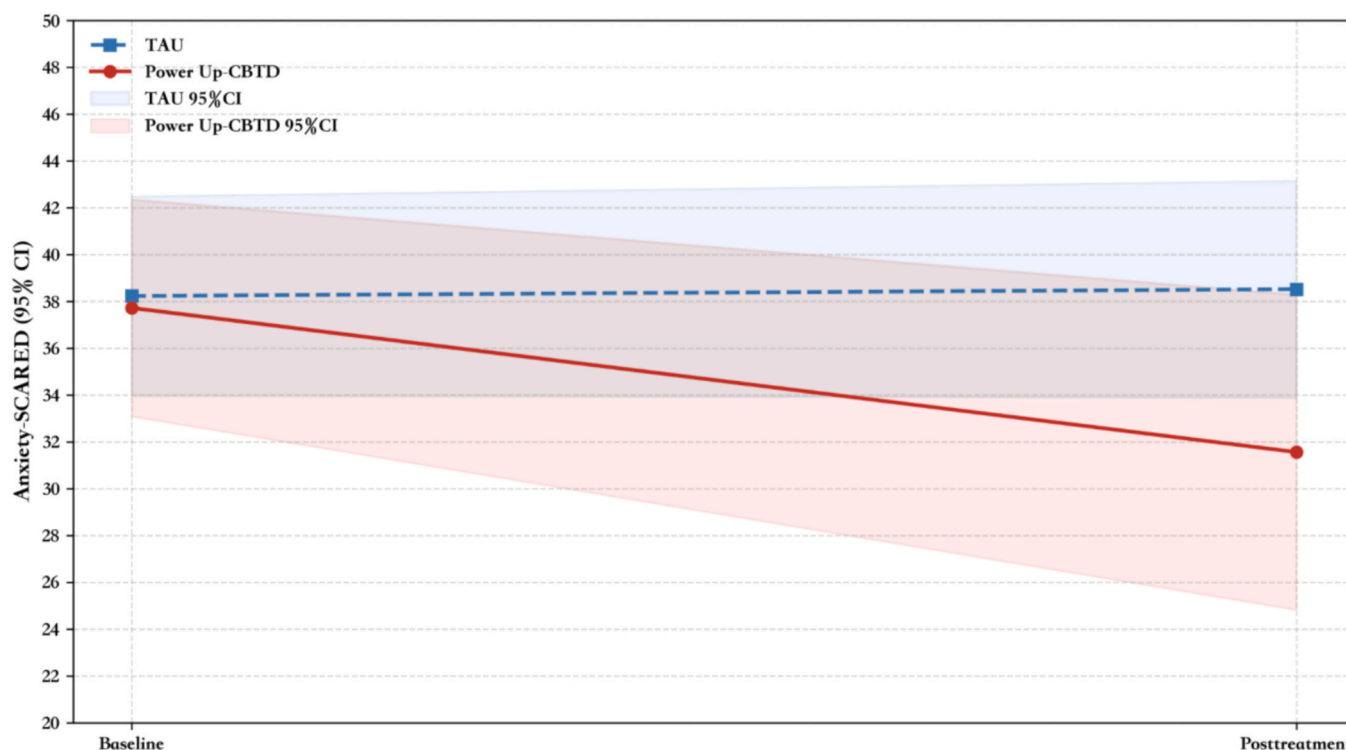


Fig. 4. Effects of the Power Up-CBTD versus TAU on SCARED scores

Notes: SCARED: Screen for Child Anxiety Related Emotional Disorders; Power Up-CBTD = power up: cognitive behavioral therapy for depressive symptoms; TAU = treatment as usual.

good adherence. The school-based format may have helped reduce some common barriers to participation, such as family constraints or scheduling difficulties (Argiros et al., 2023). Although direct comparisons should be made with caution, these rates appear favorable relative to previously reported attrition in hospital-based CBT programs (e.g., 17.5 %, Thimm and Antonsen, 2014).

Participant feedback was overwhelmingly positive: 97.5 % expressed overall satisfaction. Qualitative feedback further supported the intervention's acceptability, with students highlighting the value of interactive, peer-based learning, improved emotion regulation strategies, and the supportive, non-academic setting. These outcomes suggest the intervention was developmentally appropriate and emotionally resonant, with the interactive group format likely enhancing engagement through emotional connection, modeling, and in-session skills practice (Nardi et al., 2017).

Preliminary outcome data indicated significant reductions in depressive symptoms within the participant group, indicating the promise of this approach in preventive applications. These results are in line with earlier investigations affirming the value of CBT in mitigating mood-related challenges and delaying the onset of clinical depression (Arnberg and Öst, 2014; Nardi et al., 2017; Oud et al., 2019; Rasing et al., 2017; Yang et al., 2023), while contributing culturally relevant evidence specific to the Chinese school context.

4.1. Implementation strategies and practical implications

Beyond feasibility and clinical signals, this pilot yielded practice-facing lessons for routine school delivery. Guided by CFIR, implementation decisions targeted (a) outer setting—alignment with national policy agendas and securing political support as key facilitators of school-based delivery (Roshan et al., 2025), (b) innovation characteristics (a manualized 7-session sequence with student workbooks and visually engaging, culturally grounded activities tailored to recipients' needs), (c) inner setting (early calendar alignment with schools and a

designated room), (d) process (team formation, structured training and weekly supervision), and (e) individual characteristics—explicit role assignment (grade-level scheduler; homeroom-teacher opinion leader; two leads and two co-facilitators).

Operationally, groups were scheduled in non-academic blocks over a 7-week period, with a pre-authorized 8th session as a buffer. A familiar room was reserved, and a liaison teacher coordinated reminders and room access. Sessions followed a manual-plus-workbook sequence, consisting of a brief review, followed by new skills development, and concluding with take-home practice. Post-session debriefs were conducted to anticipate potential barriers. These features also mitigated common logistical barriers and underscore the advantages of school-based group CBTD over clinic-only or purely digital formats: colocation reduces reliance on scarce specialist personnel and caregiver-coordination demands, leverages peer processes, and decouples participation from internet and device access constraints as well as from on-line privacy risks and smartphone-use restrictions typical in Chinese schools (Xiang et al., 2018; Zamir and Wang, 2023; Zhang et al., 2025).

Effective school mental-health programs are characterized by a strong evidence base, cultural fit, and alignment with local conditions; brief, acceptable formats are more likely to be adopted in resource-constrained settings (Roshan et al., 2025). As a pilot randomized trial, this study aligns with CFIR's call for early testing of implementation readiness and rigorous intervention design, offering a foundation for future refinement and broader-scale application.

4.2. Limitations and future directions

Despite its contributions, the study is subject to certain limitations. Of 1960 students screened, approximately 465 met elevated-risk criteria, yet only 87 ultimately enrolled. This discrepancy was primarily due to a short recruitment/consent window that coincided with fixed school timetables, which substantially constrained enrollment despite high interest. As a result, some eligible students could not complete

consent procedures within the designated period. Several low-burden mitigations were piloted (early calendar alignment, a designated room and liaison teacher, and parent consent via school), but these only partly offset the narrow window. Future studies should adopt longer or rolling recruitment windows and schedule parent briefings at multiple time points to increase reach and equity.

The modest sample size in this study constrained both its statistical strength and the extent to which findings can be generalized. Follow-up data were not collected due to school transitions (e.g., students advancing to higher grades), limiting conclusions about sustained effects and relapse prevention. Although exploratory group-by-gender interactions were nonsignificant (see Sensitivity analyses), subsequent trials should pre-specify moderators and be adequately powered to detect moderation effects. Despite mitigation efforts in the acceptability interviews (separate groups, no facilitators present, neutral probes), group formats can still introduce social-influence bias. Future acceptability assessments should combine group-based, open-ended interviews (with an expanded set of neutral prompts) and brief individual interviews with group leaders to reduce peer influence and broaden perspectives.

Finally, the implementation of the intervention was carried out by graduate-level personnel, including doctoral and master's trainees, all of whom possessed relevant professional expertise. As prior research suggests that facilitator expertise is positively associated with intervention outcomes (Duagi et al., 2024), future studies should assess the effectiveness and scalability of Power Up-CBTd when delivered by trained non-specialist facilitators, such as school staff without formal clinical backgrounds. This would enhance the feasibility of broader implementation and allow more students to benefit from the program.

5. Conclusion

This pilot randomized controlled trial provides initial support for the Power Up-CBTd intervention as a manualized, brief, group-based CBT program and a developmentally appropriate school-based depression prevention approach for children and adolescents in mainland China. The intervention was feasible to deliver in school settings and was generally well received by participants, with encouraging indications of short-term symptom reduction. Nevertheless, the modest sample size, lack of follow-up, and research-led delivery limit the generalizability of these findings. Even so, the results offer practical insights into how evidence-based mental health prevention can be implemented in Chinese schools. Logistical challenges—such as narrow consent collection windows and school scheduling constraints—underscore the importance of early coordination and flexible implementation strategies in real-world contexts. Future large-scale studies should assess long-term outcomes, explore scalable delivery by trained non-specialist facilitators, and address the broader feasibility of school-based CBTd interventions through mixed-effectiveness and implementation designs.

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CRedit authorship contribution statement

Qianyun He: Writing – original draft, Visualization, Software, Methodology, Formal analysis, Conceptualization. **Jina Li:** Writing – review & editing, Data curation. **Junhe Wang:** Writing – review & editing, Validation. **Zhiyong Qu:** Writing – review & editing, Supervision, Resources, Project administration, Conceptualization.

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Declaration of competing interest

The authors declare no conflict of interest.

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Data availability

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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